Ricky Tang

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Education

University of Waterloo — Sep. 2025 - Apr. 2030
Honours Bachelor of Software Engineering (Co-op) — *GPA: 3.9/4.0*

Technical Skills

Languages: Python, TypeScript/JavaScript, SQL, Go, Ruby, Java, C#, C++, Bash, Swift, C, HTML, CSS
Frameworks: React, Next.js, Node.js, Express, FastAPI, Ruby on Rails, Spring Boot, tRPC, GraphQL, RESTful APIs, Temporal, Zod
Databases: PostgreSQL, MySQL, Redis, MongoDB, SQLite, Firebase, Supabase, SQLAlchemy, ChromaDB, Weaviate, Drizzle ORM
Infra: Git, Docker, Linux, CI/CD, AWS, GCP, Cloudflare, Vercel, Prometheus, Railway, GitHub Actions, OpenTelemetry, Sentry

Experience

Hamming AI (YC S24) | *TypeScript, Next.js, tRPC, Temporal, AWS S3, Python, LiveKit* — May 2026 - Aug. 2026
Software Engineer — *San Francisco, CA*

- Profiled 19 eval workflows by cost and routed the 2 costliest to **Haiku** after scoring parity held, cutting model spend **67%**
- Hardened call playback by copying expiring provider recordings into durable S3 storage, repairing **224K+** affected runs
- Migrated launch locking from Redis TTL expiry to durable **Postgres** leases, recovering stuck launches across 5 entry paths
- Built go-to-market pipelines producing **9,500+** attendee leads across 4 conferences from macOS app-container caches

WAT.ai x **Bindwell (YC W25)** | *Python, FastAPI, DeepEval, Weaviate, TypeScript, React* — Jan. 2026 - Sep. 2026
Software Engineer — *Waterloo, ON*

- Overhauled async BFS EPA crawler with mime sniffing and retry logic, boosting throughput **59%** and fresh PDF yield **227%**
- Engineered crash-resilient APVMA/FSANZ crawler with SQLite queue, per-prefix auto-blocking, harvesting **3900+** pages
- Architected multi-agent RAG pipeline (intent router, query architect, guardrail) over Weaviate **BM25 + vector** search
- Achieved **1.00 faithfulness** (zero hallucinations) and **1.00 answer relevancy** via DeepEval across 5 golden test questions

Develop for Good | *TypeScript, Next.js, PostgreSQL, Drizzle, Auth.js, GitHub Actions* — May 2026 - Aug. 2026
Software Engineer — *Remote*

- Led **Next.js/PostgreSQL** overhaul off legacy HTML/CSS for Illinois Swimming Officials Association, serving **69+** officials
- Architected **3-layer** authorization gating **41** route handlers, re-querying roles so admin changes revoke mid-session
- Built a self-service member portal over a searchable wiki, anonymous rulings queue, and **Cloudinary** document library

Projects

Tark – Google Earth for Game Devs | *Python, FastAPI, TypeScript, Next.js, Redis, Prometheus, Leaflet, SciPy, Numpy, Pytest*

- Developed web app turning locations into game-ready 3D meshes in **<15 seconds** (typically weeks of manual modeling)
- Processed Mapbox elevation and satellite imagery to generate terrain meshes at **45K+ triangulated faces per second**
- Extracted **2000+ building footprints** from OpenStreetMap and generated textured .obj files for Unity/Blender workflows
- Productionized with **Redis** job store, reducing latency by **30%** via asyncio workers with **98.2% accuracy** in pytest suites

BrainLattice – PDF to Obsidian Vault Generator | *Python, FastAPI, PostgreSQL, Redis, TypeScript, React, AWS, Cloudflare*

- Developed React + FastAPI web app and **npm CLI** converting PDFs into Obsidian vaults with NetworkX graph generation
- Reduced token costs by **70-80%** via global concept seeds and batched extraction replacing full-document context caching
- Accelerated processing **70%** via parallelization and embedding deduplication, generating **100+ node graphs** in **<1 minute**
- Built event-driven pipeline using **QStash + Lambda + R2** with **pgvector RAG** for on-demand notes, reaching **300+ installs**

Quota – VS Code Extension for Cost Optimization | *TypeScript, VS Code API, LangChain, Python, FastAPI, Next.js, MongoDB, GCP*

- Developed VS Code extension analyzing API/cloud/DB overhead with inline annotations and optimization suggestions
- Achieved <7s initial indexing and **<1s** refresh analysis via file hash caching, outperforming AI IDEs by **47x in speed**
- Implemented hybrid AST + regex parser detecting **2x more cost issues** than AI IDEs while saving **~50k tokens per check**
- Architected RAG-powered web sandbox with **LangChain + FAISS** delivering architectural recommendations in **<2s**

Awards

- DeltaHacks 2026: **1st Place Overall** | Hack Western 2025: **Best AI Application Built with Cloudflare**
- GenAI Genesis 2026 (Canada's largest AI hackathon): **Top 10 Overall** (out of 250+ teams)
- Hack the North 2026: **Best Use of Baseten** | Hack the North 2025: **Semi-Finalist** (top 32 out of 256 teams)